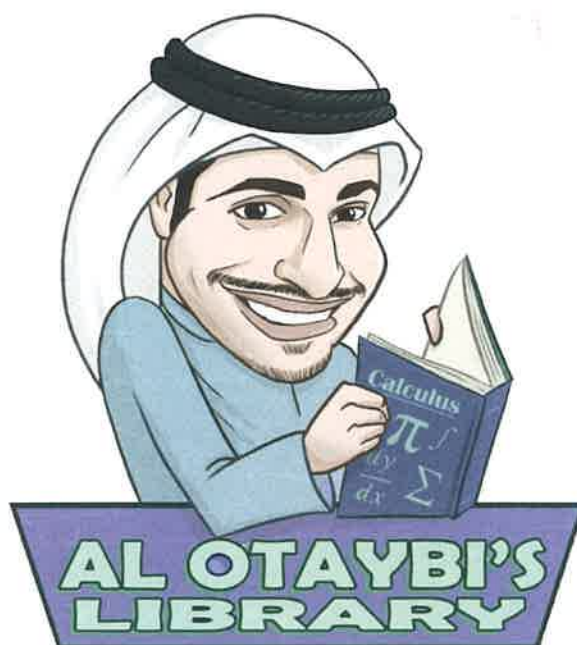


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 4.3 – 4.5 (Part b)

4.3 How Derivatives Affect the Shape of a Graph.

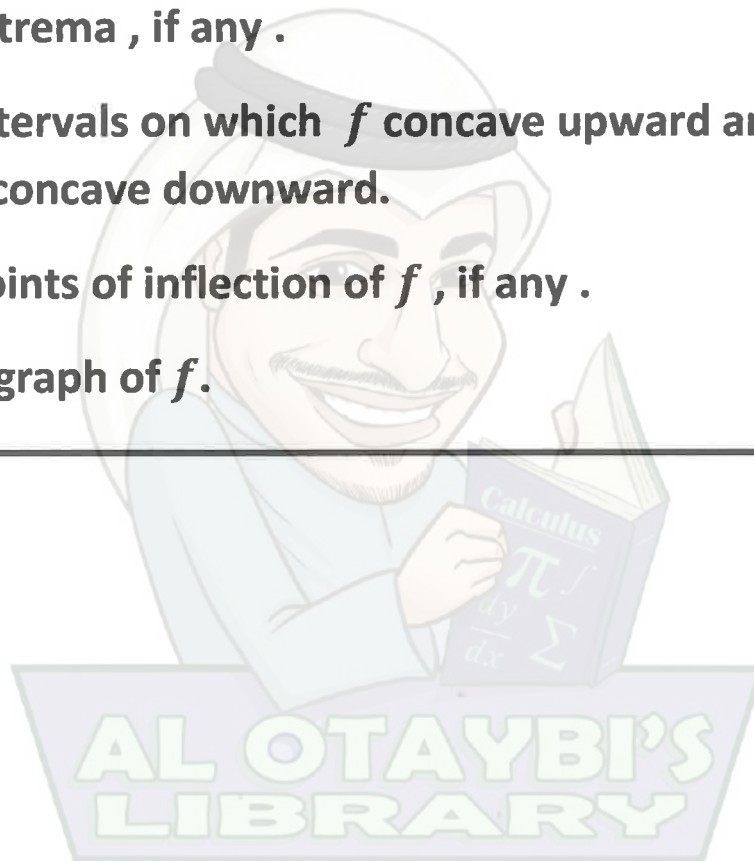
4.5 Summary of Curve Sketching.

(16)

Given that

$$f(x) = \frac{x^2}{(x-2)^2}, \quad f'(x) = \frac{-4x}{(x-2)^3} \quad \text{and} \quad f''(x) = \frac{8(x+1)}{(x-2)^4}$$

- (a) Find the vertical and horizontal asymptotes for the graph of f , if any.
- (b) Find the intervals on which f is increasing and the intervals in which f is decreasing.
- (c) Find the extrema, if any.
- (d) Find the intervals on which f concave upward and the intervals on which f is concave downward.
- (e) Find the points of inflection of f , if any.
- (f) Sketch the graph of f .



$$f(x) = \frac{x^2}{(x-2)^2}$$

Domain of $f(x)$: $x-2=0$
 $x=2$

$$\Rightarrow D_f = \mathbb{R} \setminus \{2\}$$

Horizontal asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{x^2}{(x-2)^2} = \lim_{x \rightarrow \pm\infty} \frac{x^2}{[x(1-\frac{2}{x})]^2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cancel{x^2}}{\cancel{x^2} (1-\frac{2}{x})^2} = \frac{1}{(1-0)^2} = \frac{1}{1^2} = 1$$



$\therefore$

$$\boxed{y = 1 \text{ is H.A.}}$$

Vertical asymptote

$$\underline{x=2}$$

$$\lim_{x \rightarrow 2^+} \frac{x^2}{(x-2)^2} = \frac{4}{0^+} = +\infty$$

$$\lim_{x \rightarrow 2^-} \frac{x^2}{(x-2)^2} = \frac{4}{0^+} = +\infty$$



$\therefore$

$$\boxed{x = 2 \text{ is V.A.}}$$

$$f'(x) = \frac{-4x}{(x-2)^3}$$

$= 0$

DNE

$$\begin{aligned} -4x &= 0 \\ \div (-4) \end{aligned}$$

$x = 0$
 $\in \text{Domain}$

$$x - 2 = 0$$

$x = 2$
 $\notin \text{Domain}$



$f'(x)$	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$\frac{-4x}{(x-2)^3}$	+	-	-
	-	+	-

$\ominus$

$\oplus$

$\ominus$

$\therefore f(x)$ is increasing on $(0, 2)$
 $\therefore f(x)$ is decreasing on $(-\infty, 0)$ and $(2, \infty)$

at $x = 0$ there is Local minimum

$$y = \frac{0}{(0-2)^2} = 0$$

$\Rightarrow$

$(0, 0)$ is Local minimum

$$f''(x) = \frac{8(x+1)}{(x-2)^4}$$

$= 0$

$$\begin{aligned} 8(x+1) &= 0 \\ \div 8 \quad x+1 &= 0 \\ x &= -1 \end{aligned}$$

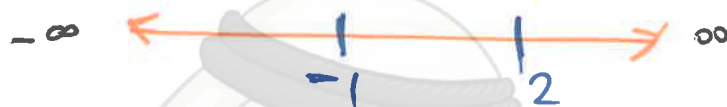
$\in \text{Domain}$

DNE

$$x-2=0$$

$$x=2$$

$\notin \text{Domain}$



$f''(x)$	$(-\infty, -1)$	$(-1, 2)$	$(2, \infty)$
$\frac{8(x+1)}{(x-2)^4}$	-	+	+
	⊖	⊕	⊕
	∩	∪	∪

$\therefore f(x)$ is concave up on $(-1, 2)$ and $(2, \infty)$
 $f(x)$ is concave downward on $(-\infty, -1)$

at $x = -1$ there is inflection point

$$y = \frac{(-1)^2}{(-1-2)^2} = \frac{1}{(-3)^2} = \frac{1}{9}$$

$\Rightarrow \therefore (-1, \frac{1}{9})$ is inflection point

Key points

Intersection point
with HA

Random
point

$$1 = \frac{x^2}{(x-2)^2}$$

$$* (x-2)^2$$

$$(x-2)^2 = x^2$$

$$x^2 - 4x + 4 = x^2$$

$$\cancel{x^2} - 4x + 4 - \cancel{x^2} = 0$$

$$4x = 4$$

$$\div 4$$

$$x = 1$$

$\Rightarrow$

$$(1, 1)$$

H.A

$$\begin{array}{c|c} -2 & 3 \\ \hline & 2 \end{array}$$

$$\text{at } x = -2$$

$$y = \frac{4}{(-4)^2} = \frac{4}{16} = \frac{1}{4}$$

$\Rightarrow$

$$(-2, \frac{1}{4})$$

$$\text{at } x = 3$$

$$y = \frac{9}{(3-2)^2} = \frac{9}{1} = 9$$

$\Rightarrow$

$$(3, 9)$$

$$D_f = \{R \setminus \{2\}\}$$

$$H.A \rightarrow y=1$$

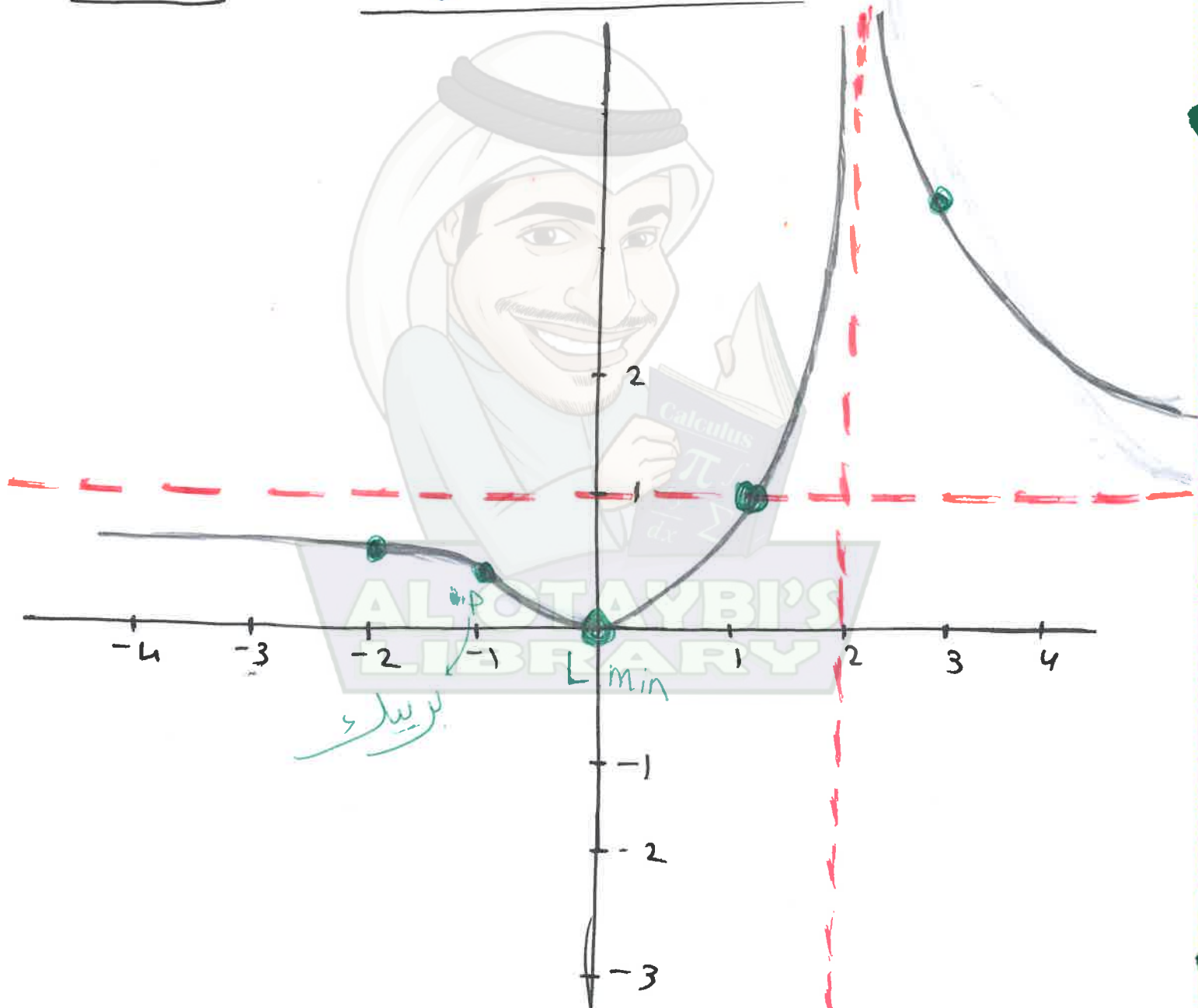
$$V.A \rightarrow x=2 \quad \begin{matrix} \nearrow \oplus +\infty \\ \searrow \ominus +\infty \end{matrix}$$

$$(0,0) \text{ L.min}$$

$$\cup (-1, 2) \quad (2, \infty) \quad \cap (-\infty, -1)$$

$$(-1, \frac{1}{9}) \text{ inf. point.}$$

$$\boxed{(1, 1)} \quad (-2, \frac{1}{4}) \quad (3, 9)$$



Given that $f(x) = \frac{3 - x}{(x - 1)^2}$

(a) Find the vertical and horizontal asymptotes for the graph of f , if any.

Given that $f'(x) = \frac{x - 5}{(x - 1)^3}$

(b) Find the intervals on which f is increasing and the intervals in which f is decreasing.

(c) Find the local maximum and local minimum values of f , if any .

Given that $f''(x) = \frac{2(7 - x)}{(x - 1)^4}$

(d) Find the intervals on which f concave upward and the intervals on which f is concave downward.

(e) Find the points of inflection of f , if any .

(f) Sketch the graph of f .

(g) Find the absolute maximum and absolute minimum of f on $[2, 4]$



هناك سؤال

مخشوش !!



$$f(x) = \frac{3-x}{(x-1)^2}$$

Domain of $f(x)$: $x-1=0 \Rightarrow x=1$

$$\Rightarrow D_f = \mathbb{R} \setminus \{1\}$$

Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{3-x}{(x-1)^2} = \lim_{x \rightarrow \pm\infty} \frac{x \left[\frac{3}{x} - 1 \right]}{\left[x \left(1 - \frac{1}{x} \right) \right]^2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cancel{x} \left[\frac{3}{x} - 1 \right]}{\cancel{x^2} \left(1 - \frac{1}{x} \right)^2} = \lim_{x \rightarrow \pm\infty} \frac{\frac{3}{x} - 1}{x \left(1 - \frac{1}{x} \right)^2}$$

$$\Rightarrow \boxed{y=0 \text{ is H.A.}} = \frac{0-1}{\pm\infty(1-0)^2} = \frac{-1}{\pm\infty} = 0$$

Vertical asymptote

$x=1$

$$\lim_{x \rightarrow 1^+} \frac{3-x}{(x-1)^2} = \frac{2}{0^+} = +\infty$$

دائماً $+$ لأن المقام تربيع

$$\lim_{x \rightarrow 1^-} \frac{3-x}{(x-1)^2} = \frac{2}{0^+} = +\infty$$

$$\Rightarrow \boxed{\therefore x=1 \text{ is V.A.}}$$

$$f'(x) = \frac{x-5}{(x-1)^3}$$

$$= 0$$

$$x-5=0$$

$$x=5$$

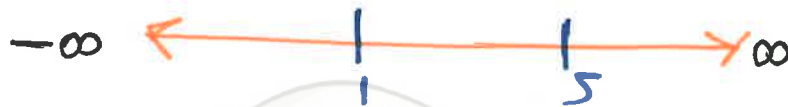
∈ Domain

DNE

$$x-1=0$$

$$x=1$$

∉ Domain



$f'(x)$	$(-\infty, 1)$	$(1, 5)$	$(5, \infty)$
$\frac{x-5}{(x-1)^3}$	-	-	+
	-	+	-

⊕

⊖

⊕

∴ $f(x)$ is increasing on $(-\infty, 1)$ and $(5, \infty)$

∴ $f(x)$ is decreasing on $(1, 5)$

at $x=5$ there is Local minimum

$$y = \frac{3-5}{(5-1)^2} = \frac{-2}{(4)^2} = \frac{-2}{16} = -\frac{1}{8}$$

⇒

$(5, -\frac{1}{8})$ is Local minimum

$$f''(x) = \frac{2(7-x)}{(x-1)^4}$$

$= 0$

$$2(7-x) = 0$$

$\div 2$

$$7-x = 0$$

$$x = 7$$

$\in \text{Domain}$

DNE

$$x-1=0$$

$$x=1$$

$\notin \text{Domain}$



$f''(x)$	$(-\infty, 1)$	$(1, 7)$	$(7, \infty)$
$\frac{2(7-x)}{(x-1)^4}$	+	+	-
	+	+	+
	$\oplus$	$\oplus$	$\ominus$
	$\cup$	$\cup$	$\cap$

- $\therefore f(x)$ is concave up on $(-\infty, 1)$ and $(1, 7)$
- $\therefore f(x)$ is concave downward on $(7, \infty)$

at $x=7$ there is inflection point

$$y = \frac{3-7}{(7-1)^2} = \frac{-4}{(6)^2} = \frac{-4}{36} = -\frac{1}{9}$$

$\Rightarrow (7, -\frac{1}{9})$ is inflection point

Key points

Intersection point with HA

Random point

$$0 = \frac{3-x}{(x-1)^2}$$

$$* (x-1)^2$$

$$0 = 3 - x$$

$$x = 3$$

$$(3, 0)$$

H.A

at $x=0$

$$y = \frac{3-0}{(0-1)^2} = \frac{3}{(-1)^2} = 3$$

$$\Rightarrow (0, 3)$$

at $x=2$

$$y = \frac{3-2}{(2-1)^2} = \frac{1}{1^2} = 1$$

$$\Rightarrow (2, 1)$$

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LIBRARY

$$HA \rightarrow y=0$$

$$V.A \rightarrow x=1$$

$$\begin{matrix} \oplus & +\infty \\ \ominus & +\infty \end{matrix}$$



$$(-\infty, 1) \quad (1, 7)$$



$$(7, \infty)$$

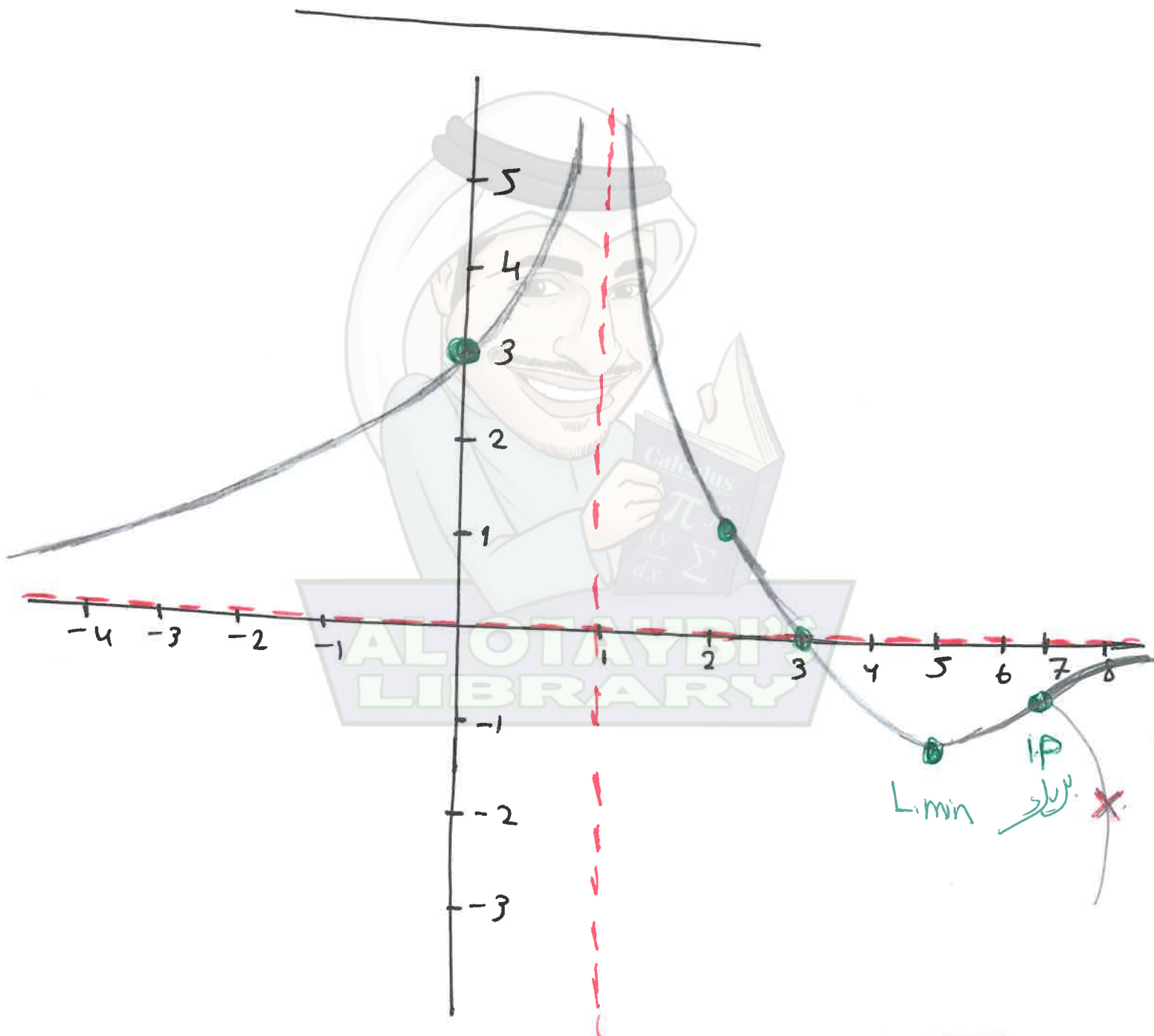
$$(5, -\frac{1}{8}) \text{ L.min}$$

$$(7, -\frac{1}{9}) \text{ inf. point}$$

$$(3, 0)$$

$$(0, 3)$$

$$(2, 1)$$



سؤال مَشَوَسْ (g)

$$f'(x) = \frac{x-5}{(x-1)^3} \quad [2, 4]$$

$= 0$

$$x - 5 = 0$$

$$x = 5$$

$\notin (2, 4)$

DNE

$$x - 1 = 0$$

$$x = 1$$

$\notin (2, 4)$

$$f(2) = \frac{3-2}{(2-1)^2} = \frac{1}{1^2} = 1$$

$$f(4) = \frac{3-4}{(4-1)^2} = \frac{-1}{3^2} = -\frac{1}{9}$$

∴ absolute maximum is 1

absolute minimum is $-\frac{1}{9}$

Given that $f(x) = \frac{x}{x^2 - 1}$

(a) Find the vertical and horizontal asymptotes for the graph of f , if any.

Given that $f'(x) = -\frac{x^2 + 1}{(x^2 - 1)^2}$

(b) Find the intervals on which f is increasing and the intervals in which f is decreasing.

(c) Find the local maximum and local minimum values of f , if any.

Given that $f''(x) = \frac{2x(x^2 + 3)}{(x^2 - 1)^3}$

(d) Find the intervals on which f concave upward and the intervals on which f is concave downward.

(e) Find the points of inflection of f , if any.

(f) Find the x-intersect and y-intercept and study the symmetry.

(g) Sketch the graph of f .



هناك سؤال
مخشوش !!



$$f(x) = \frac{x}{x^2 - 1}$$

Domain of $f(x)$: $x^2 - 1 = 0$
 $x^2 = 1$
 $x = -1 \quad x = 1$

$$\Rightarrow D_f = \mathbb{R} \setminus \{\pm 1\}$$

Horizontal asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{x}{x^2 - 1} = \lim_{x \rightarrow \pm\infty} \frac{x}{x^2 [1 - \frac{1}{x^2}]}$$

$$\lim_{x \rightarrow \pm\infty} \frac{1}{x [1 - \frac{1}{x^2}]} = \frac{1}{\pm\infty [1 - 0]} = \frac{1}{\pm\infty} = 0$$

$$\Rightarrow \boxed{y = 0 \text{ is H.A.}}$$

Vertical asymptote

$x = 1$ $\cdot \lim_{x \rightarrow 1^+} \frac{x}{(x-1)(x+1)} = \frac{1}{0^+} = +\infty$

$\lim_{x \rightarrow 1^-} \frac{x}{(x-1)(x+1)} = \frac{1}{0^-} = -\infty$

$$\Rightarrow \therefore \boxed{x = 1 \text{ is V.A.}}$$

$x = -1$ $\cdot \lim_{x \rightarrow -1^+} \frac{x}{(x-1)(x+1)} = \frac{-1}{0^-} = +\infty$

$\lim_{x \rightarrow -1^-} \frac{x}{(x-1)(x+1)} = \frac{-1}{0^+} = -\infty$

$$\Rightarrow \boxed{x = -1 \text{ is V.A.}}$$

$$f'(x) = - \frac{x^2 + 1}{(x^2 - 1)^2}$$

$$= 0$$

$$-(x^2 + 1) = 0$$

$$\div (-1)$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

None

DNE

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

$\notin$ Domain



$f'(x)$	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$-\frac{(x^2 + 1)}{(x^2 - 1)^2}$	$-$	$-$	$-$
	$+$	$+$	$+$

$\ominus$

$\ominus$

$\ominus$

$\therefore f(x)$ is decreasing on $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$

$\therefore$ No interval of increasing.

$\Rightarrow \therefore$ No Local Extrema.

$$f''(x) = \frac{2x(x^2 + 3)}{(x^2 - 1)^3}$$

$= 0$

$$2x(x^2 + 3) = 0$$

$$2x = 0$$

$$\div 2$$

$$x = 0$$

$\in \text{Domain}$

$$x^2 + 3 = 0$$

$$x^2 = -3$$

None

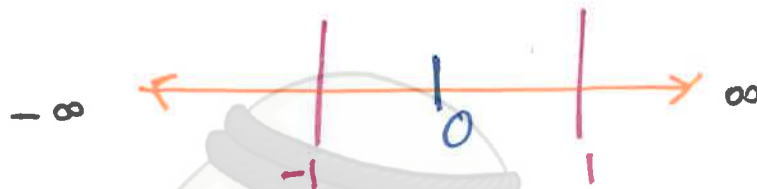
$\neq$

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

$\notin \text{Domain}$



$f''(x)$	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
$\frac{2x(x^2+3)}{(x^2-1)^3}$	-	-	+	+
	+	-	-	+
	$\ominus$	$\oplus$	$\ominus$	$\oplus$
	$\cap$	$\cup$	$\cap$	$\cup$

$\therefore f(x)$ is concave up on $(-1, 0)$ and $(1, \infty)$

$\therefore f(x)$ is concave downward $(-\infty, -1)$ and $(0, 1)$

at $x=0$ there is inflection point

$$y = \frac{0}{0-1} = \frac{0}{-1} = 0$$

$\therefore (0, 0)$ is inflection point

Key points

intersection point with H.A

H.A $\rightarrow 0 = \frac{x}{x^2-1}$
 $\times (x^2-1)$

$$0 = x$$

فاله داي نطلع
 لان

H.A $\rightarrow$
 $P(0, 0)$

Random points

$$-2 \left| \begin{array}{c} \cancel{x} \\ \frac{1}{2} \end{array} \right| \begin{array}{c} 2 \\ 1 \end{array}$$

at $x = -2 \rightarrow y = \frac{-2}{4-1} = -\frac{2}{3}$

$$\Rightarrow (-2, -\frac{2}{3})$$

at $x = \frac{1}{2} \rightarrow y = \frac{\frac{1}{2}}{\frac{1}{4}-1} = \frac{\frac{1}{2}}{\frac{1}{4}-\frac{4}{4}}$

$$= \frac{\frac{1}{2}}{-\frac{3}{4}} = \frac{\cancel{4}^2}{-3(\cancel{2})} = -\frac{2}{3}$$

$$\Rightarrow (\frac{1}{2}, -\frac{2}{3})$$

at $x = 2 \rightarrow y = \frac{2}{4-1} = \frac{2}{3}$

$$\Rightarrow (2, \frac{2}{3})$$

$$D_f = \mathbb{R} \setminus \{\pm 1\}$$

$$HA \rightarrow y=0$$

$$V.A \begin{cases} X=1 \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases} \\ X=-1 \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases} \end{cases}$$

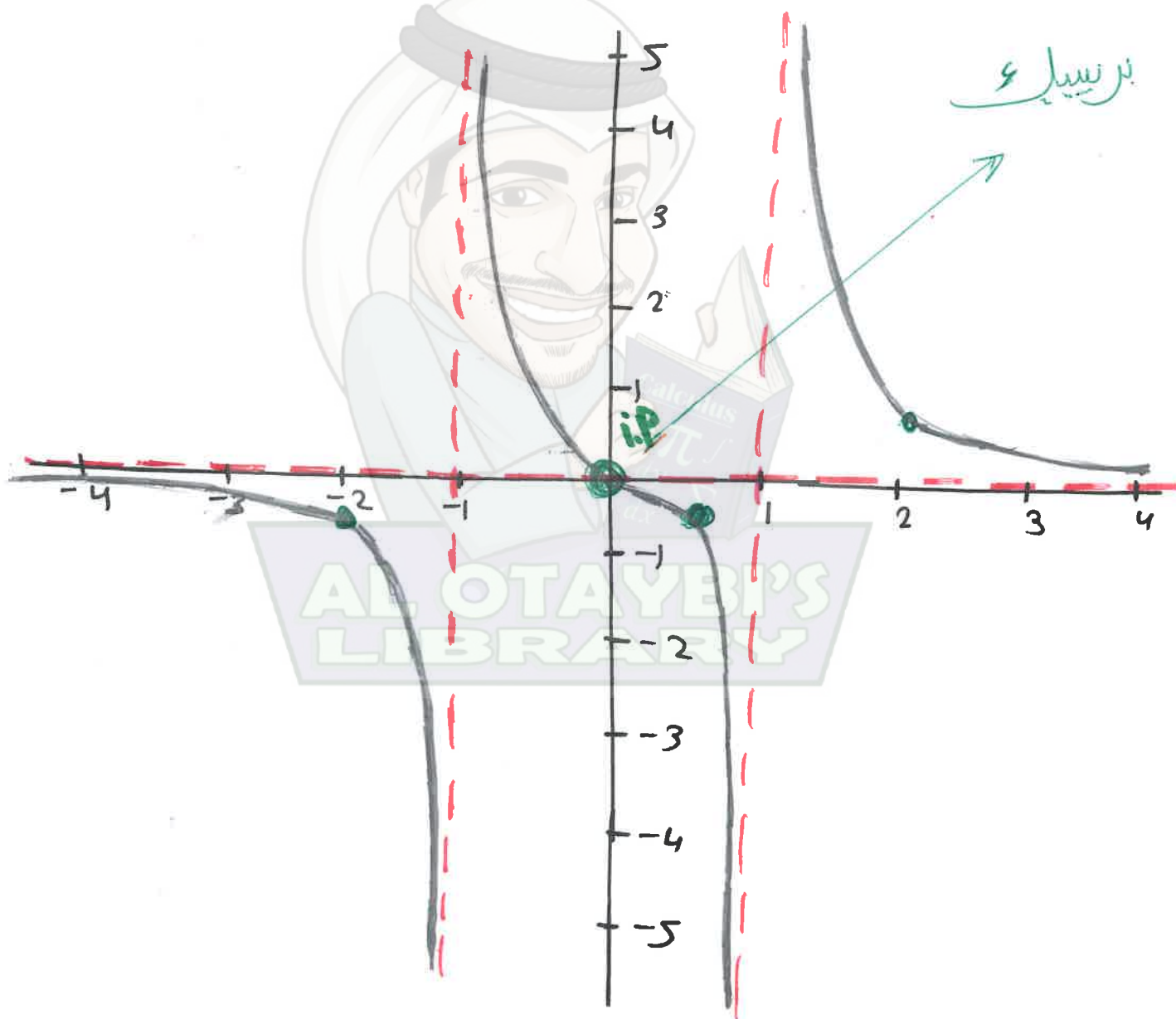
$$\cap \begin{pmatrix} -\infty & -1 \\ 0 & 1 \end{pmatrix}$$

$$\cup \begin{pmatrix} -1 & 0 \\ 1 & \infty \end{pmatrix}$$

$(0,0)$ inf. point

$$\boxed{(0,0)} \quad (-2, -\frac{2}{3}) \quad (\frac{1}{2}, -\frac{2}{3}) \quad (2, \frac{2}{3})$$

اقل من 1



سؤال مختصر (ف)

$$y = \frac{x}{x^2 - 1}$$

x-intercept

y-intercept

$$0 = \frac{x}{x^2 - 1}$$

$$* (x^2 - 1)$$

$$0 = x$$

$$(0, 0)$$

$$y = \frac{0}{0 - 1} = \frac{0}{-1} = 0$$

$$(0, 0)$$

$$f(x) = \frac{x}{x^2 - 1}$$

$$f(-x) = \frac{(-x)}{(-x)^2 - 1} = \frac{-x}{x^2 - 1}$$

$$= -f(x)$$

$$\therefore f(-x) = -f(x)$$

$\Rightarrow \therefore f(x)$ is symmetry about origin

$\therefore f(x)$ is odd function



Try to solve !!



Given that

$$f(x) = \frac{2x^2 - 2x + 1}{x^2}, \quad f'(x) = \frac{2x - 2}{x^3} \quad \text{and} \quad f''(x) = \frac{6 - 4x}{x^4}$$

- Find the vertical and horizontal asymptotes for the graph of f , if any.
- Find the intervals on which f is increasing and the intervals in which f is decreasing.
- Find the local maximum and local minimum values of f , if any.
- Find the intervals on which f concave upward and the intervals on which f is concave downward.
- Find the points of inflection of f , if any.
- Sketch the graph of f .

Ans.

HA $\rightarrow y = 2$

V.A $\rightarrow x = 0$

int. of inc. $(-\infty, 0) \quad (1, \infty)$

int. of dec. $(0, 1)$

$(1, 1)$ L. min

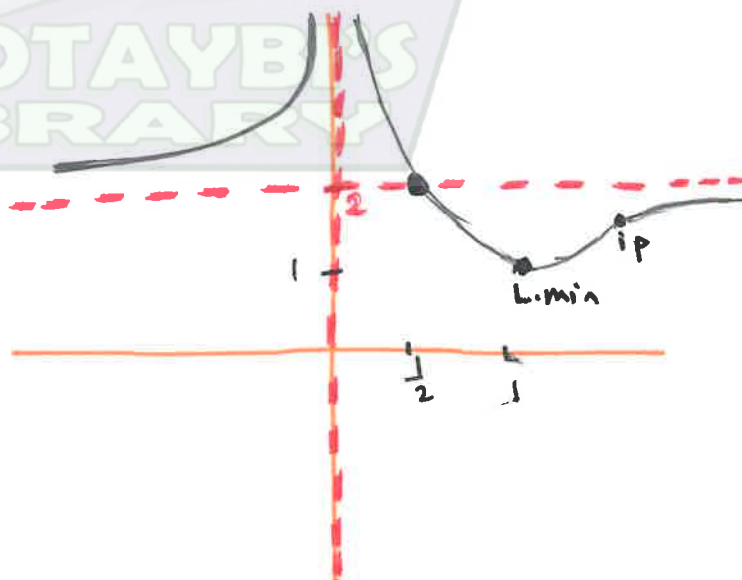
$\cup (-\infty, 0) \quad (0, \frac{3}{2})$

$\cap (\frac{3}{2}, \infty)$

$(\frac{3}{2}, \frac{10}{9})$ i.p

int. with H.A

$(\frac{3}{2}, 2)$



Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi

Let $f(x) = \frac{1}{(e^x - 1)^2}$

a) Find the vertical and horizontal asymptotes, if any.

b) Given that $f'(x) = \frac{-2e^x}{(e^x - 1)^3}$

i) Find the intervals of increasing or decreasing.

ii) Find the local maximum and minimum, if any.

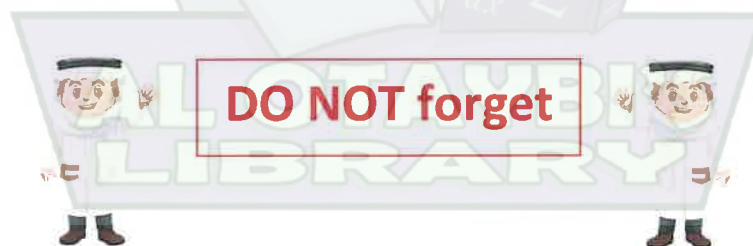
c) Given that $f''(x) = \frac{2e^x(2e^x + 1)}{(e^x - 1)^4}$

i) Find the intervals of concave up or down, if any.

ii) Find the points of inflection, if any.

d) Sketch the graph of f .

Past exam



On Limit

$$e^{-\infty} = 0, \quad e^{\infty} = \infty, \quad e^0 = 1$$

$$f(x) = \frac{1}{(e^x - 1)^2}$$

Domain:

$$\begin{aligned} e^x - 1 &= 0 \\ e^x &= 1 \\ \ln e^x &= \ln 1 \\ x &= 0 \\ \mathbb{R} \setminus \{0\} \end{aligned}$$

HA

$$\lim_{x \rightarrow \infty} \frac{1}{(e^x - 1)^2} = \frac{1}{\infty} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{1}{(e^x - 1)^2} = \frac{1}{(0 - 1)^2} = \frac{1}{1} = 1$$

∴

$y = 0$ $y = 1$ are HAs

V.A

at $x = 0$

$$\lim_{x \rightarrow 0^+} \frac{1}{(e^x - 1)^2} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{(e^x - 1)^2} = \frac{1}{0^+} = +\infty$$

∴

$x = 0$ is V.A

$$f'(x) = \frac{-2e^x}{(e^x - 1)^3}$$

$$= 0$$

$$-2e^x = 0$$

$$e^x = 0$$

None

DNE

$$e^x - 1 = 0$$

$$e^x = 1$$

$$\ln e^x = \ln 1$$

$$x = 0$$

$\notin \text{Dom}$



$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{-2e^x}{(e^x - 1)^3}$	-	-

$\oplus$

$\ominus$

$\therefore f(x)$ is increasing on $(-\infty, 0)$
 $f(x)$ is decreasing on $(0, \infty)$

No Local Extrema.

$$f''(x) = \frac{2e^x(2e^x+1)}{(e^x-1)^4}$$

$= 0$

$$2e^x(2e^x+1) = 0$$

$$2e^x = 0$$

$$e^x = 0$$

None

$$2e^x + 1 = 0$$

$$2e^x = -1$$

None

DNE

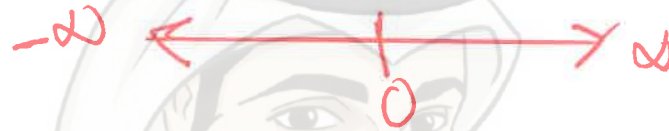
$$e^x - 1 = 0$$

$$e^x = 1$$

$$\ln e^x = \ln 1$$

$$x = 0$$

$\notin \text{Dom.}$



$f''(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{2e^x(2e^x+1)}{(e^x-1)^4}$	+	+

(+)

(+)

$\therefore f(x)$ is concave up on $(-\infty, 0) \cup (0, \infty)$
 No interval of concave down

Key points

intersection points with HA

$$0 = \frac{1}{(e^x - 1)^2}$$

$$0 = 1$$

None

$$1 = \frac{1}{(e^x - 1)^2}$$

$$(e^x - 1)^2 = 1$$

$$e^{2x} - 2e^x + 1 = 1$$

$$e^{2x} - 2e^x = 0$$

$$e^x (e^x - 2) = 0$$

$$e^x = 0 \quad e^x = 2$$

None

$$\ln e^x = \ln 2$$

$$x = \ln 2$$

$$\boxed{(\ln 2, 1)}$$

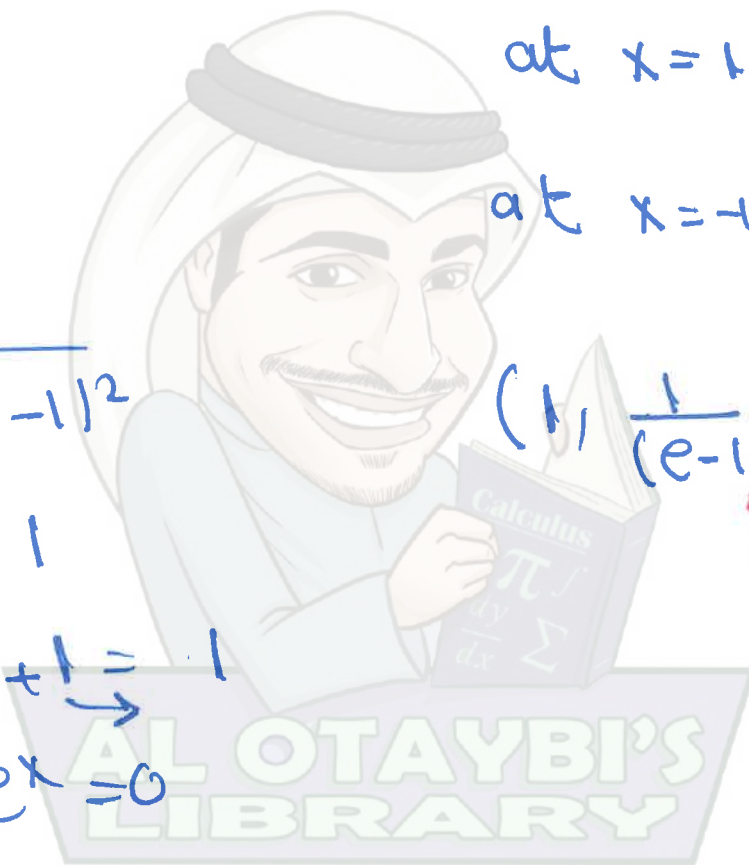
$$\begin{array}{c} -1 \\ | \\ 0 \\ | \\ 1 \end{array}$$

$$\text{at } x=1 \rightarrow y = \frac{1}{(e-1)^2}$$

$$\text{at } x=-1 \rightarrow y = \frac{1}{(e^{-1}-1)^2}$$

$$(1, \frac{1}{(e-1)^2}), (-1, \frac{1}{(e^{-1}-1)^2})$$

بالتالي

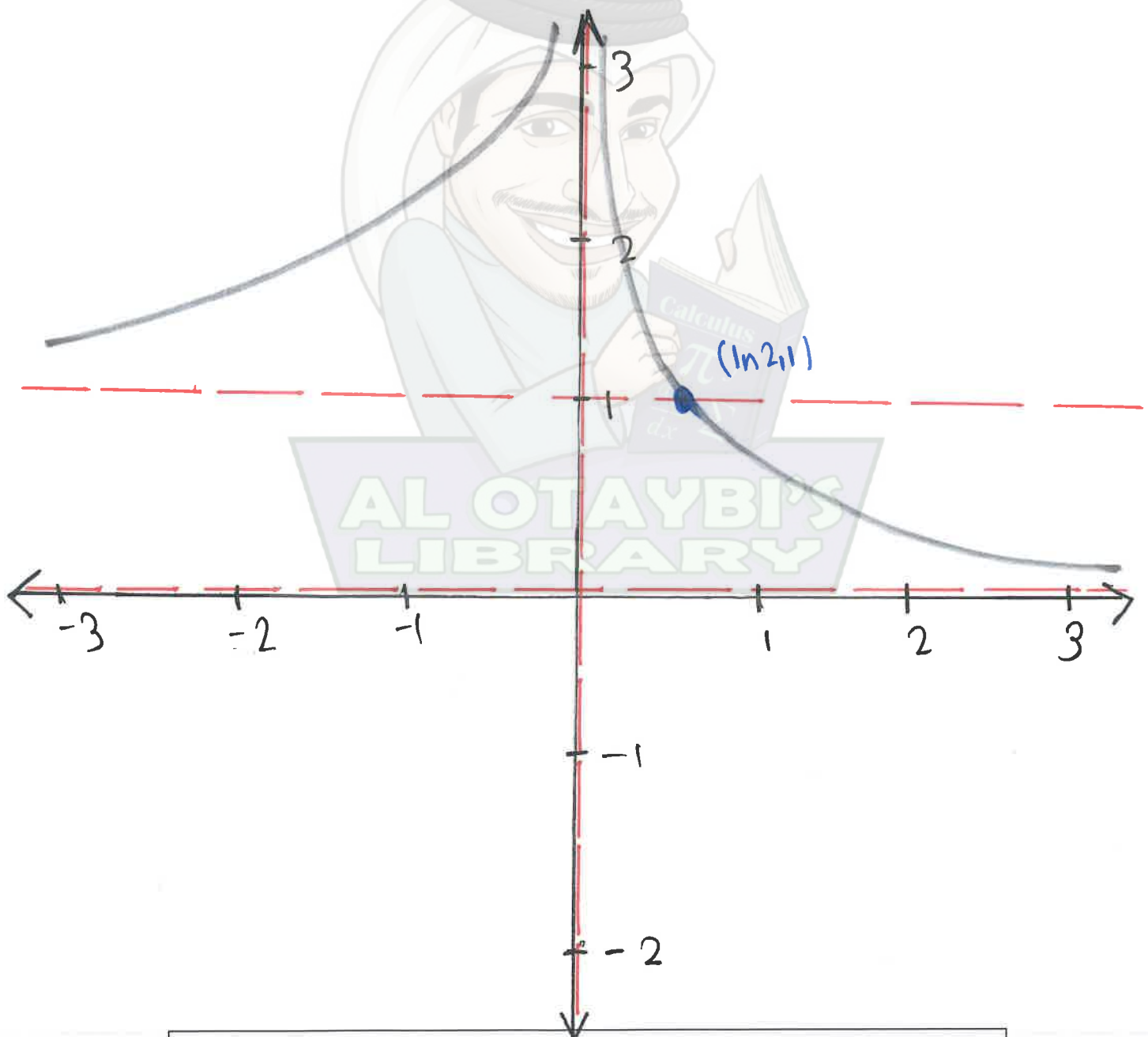


$$\begin{aligned}
 &HA \begin{cases} \rightarrow y=0 \\ \rightarrow y=1 \end{cases} \\
 &V.A \rightarrow x=0 \begin{cases} \rightarrow \oplus +\infty \\ \rightarrow \ominus +\infty \end{cases}
 \end{aligned}$$

$$U \quad (-\infty, 0) \quad (0, \infty)$$

$$(1 \ln 2, 1)$$

$$\left(1, \frac{1}{(e-1)^2}\right) \quad \left(-1, \frac{1}{(e^{-1}-1)^2}\right)$$



Let $f(x) = \frac{e^x}{e^x - 1}$

a) Study the symmetry of the curve and find the x – *intercepts* and y – *intercepts* , if any.

b) Find the vertical and horizontal asymptotes for the graph of f , if any.

c) Given that $f'(x) = \frac{-e^x}{(e^x - 1)^2}$

i) Find the intervals on which f is increasing and the intervals on which f is decreasing.

ii) Find the local maximum and minimum values of f , if any.

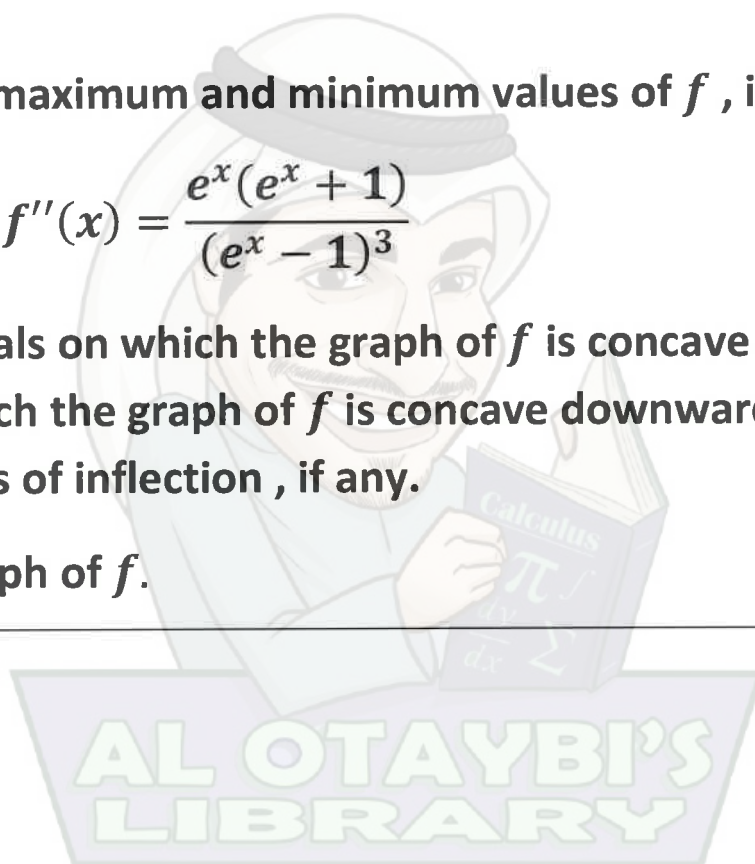
d) Given that $f''(x) = \frac{e^x(e^x + 1)}{(e^x - 1)^3}$

i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward.

ii) Find the points of inflection , if any.

e) Sketch the graph of f .

Past exam



DO NOT forget



On Limit

$$e^{-\infty} = 0 \quad , \quad e^{\infty} = \infty \quad , \quad e^0 = 1$$

Domain of $f(x)$:

$$e^x - 1 = 0$$

$$e^x = 1$$

$$\ln e^x = \ln 1$$

$$x = 0$$

$$\Rightarrow \therefore D_f = \mathbb{R} \setminus \{0\}$$

Horizontal asymptote

$$\lim_{x \rightarrow \infty} \frac{e^x}{e^x - 1} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{e^x \left[1 - \frac{1}{e^x} \right]} = \frac{1}{1 - \frac{1}{\infty}} = \frac{1}{1 - 0} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{e^x}{e^x - 1} = \frac{e^{-\infty}}{e^{-\infty} - 1} = \frac{0}{0 - 1} = 0$$

$$\Rightarrow \therefore \boxed{y=1} \text{ and } \boxed{y=0} \text{ are HA's}$$

Vertical asymptote

$$\lim_{x \rightarrow 0^+} \frac{e^x}{e^x - 1} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{e^x}{e^x - 1} = \frac{1}{0^-} = -\infty$$

$$\Rightarrow \therefore \boxed{x=0} \text{ is V.A}$$

$$f'(x) = \frac{-e^x}{(e^x - 1)^2}$$



= 0

$$-e^x = 0$$

$$\div (-1) \quad e^x = 0$$

None

undefined

$$e^x - 1 = 0$$

$$e^x = 1$$

$$\ln e^x = \ln 1$$

$$x = 0$$

$\notin D_f$



$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{-e^x}{(e^x - 1)^2}$	$\begin{matrix} - \\ + \\ \ominus \end{matrix}$	$\begin{matrix} - \\ + \\ \ominus \end{matrix}$

∴ interval of decreasing are $(-\infty, 0)$ and $(0, \infty)$
No interval of increasing

No Local extrema.

$$f''(x) = \frac{e^x(e^x+1)}{(e^x-1)^3}$$

= 0

$$e^x(e^x+1) = 0$$

$$e^x = 0$$

None

$$e^x+1 = 0$$

$$e^x = -1$$

None

undefined

$$e^x - 1 = 0$$

$$e^x = 1$$

$$\ln e^x = \ln 1$$

$$x = 0$$

$\notin D_f$



$f''(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{e^x(e^x+1)}{(e^x-1)^3}$	+	+
	-	+
	$\ominus$	$\oplus$
	$\cap$	$\cup$

∴ The interval of concave upward on $(0, \infty)$
The interval of concave downward on $(-\infty, 0)$

No inflection point

Key points

Intersection point with HA

Random points

$$1 = \frac{e^x}{e^x - 1}$$

$$e^x - 1 = e^x$$

$$\xrightarrow{\quad} -1 = 0$$

None

$$0 = \frac{e^x}{e^x - 1}$$

$$0 = e^x$$

None

$$\frac{-1}{0} \quad \frac{1}{0}$$

$$\text{at } x = -1 \rightarrow y = \frac{e^{-1}}{e^{-1} - 1}$$

$$\text{at } x = 1 \rightarrow y = \frac{e}{e - 1}$$

$$\left(-1, \frac{e^{-1}}{e^{-1} - 1}\right), \left(1, \frac{e}{e - 1}\right)$$

* Intersection *

x-axis

$$0 = \frac{e^x}{e^x - 1}$$

$$*(e^x - 1)$$

$$0 = e^x$$

None

y-axis

$$y = \frac{e^0}{e^0 - 1} = \frac{1}{0} \text{ (undefined)}$$

None

$$f(-x) = \frac{e^{-x}}{e^{-x} - 1}$$

$\neq$

$$f(x)$$

$$-f(x)$$

$\therefore$ No Symmetry.

$$D_f = \mathbb{R} \setminus \{0\}$$

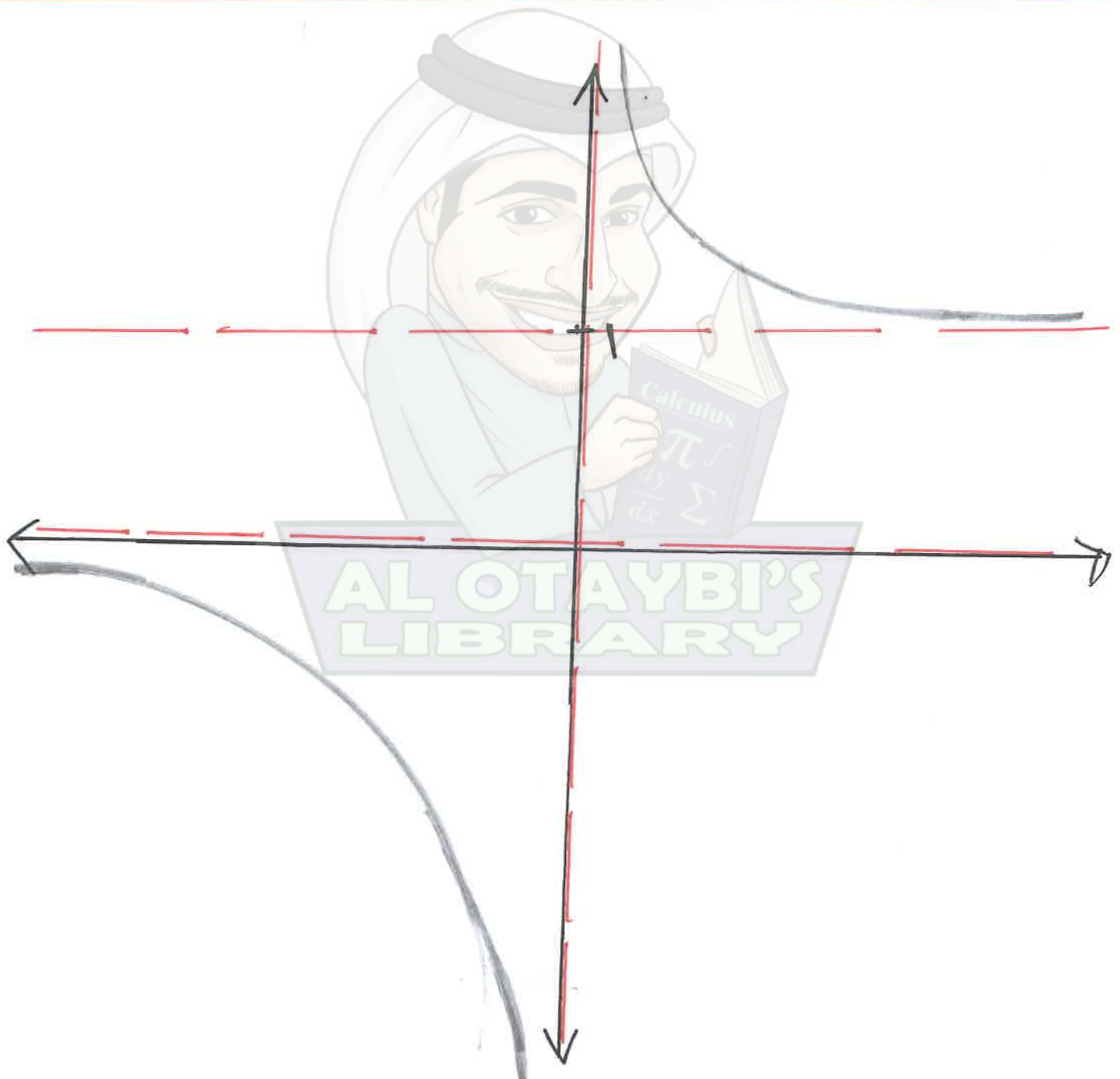
$$H.A. \begin{cases} y=1 \\ y=0 \end{cases}$$

$$V.A \rightarrow X=0 \begin{cases} \oplus +\infty \\ \ominus -\infty \end{cases}$$

$$\cup (0, \infty)$$

$$\cap (-\infty, 0)$$

$$\left(-1, \frac{e^{-1}}{e^{-1}-1}\right) \quad \left(1, \frac{e}{e-1}\right)$$



Let $f(x) = e^{-x^2}$

- a) Find the vertical and horizontal asymptotes, if any.
- b) Find the intervals of increasing or decreasing.
- c) Find the local maximum and minimum, if any.
- d) Find the intervals of concave up or down, if any.
- e) Find the points of inflection, if any.
- f) Sketch the graph of f .

Domain of $f(x) = \mathbb{R}$

Horizontal asymptote

$$\lim_{x \rightarrow \infty} e^{-x^2} = e^{-\infty} = 0$$

$$\lim_{x \rightarrow -\infty} e^{-x^2} = e^{-\infty} = 0$$

$\Rightarrow \therefore y=0$ is HA

Vertical asymptote

No Vertical asymptote.

$$f'(x) = e^{-x^2} (-2x)$$

$$= -2x e^{-x^2}$$

$$= 0$$

$$-2x e^{-x^2} = 0$$

$$-2x = 0 \quad e^{-x^2} = 0$$

$$x = 0 \in \text{Domain No Ne}$$

DNE

No Ne

$$-\infty \leftarrow 0 \rightarrow \infty$$

$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$-2x e^{-x^2}$	+	-

∞ interval of increasing $(-\infty, 0)$
 interval of decreasing $(0, \infty)$

at $x = 0 \rightarrow$ Local maximum

$$y = e^0 = 1$$

$\Rightarrow (0, 1)$ is L.max

$$\begin{aligned}
 f''(x) &= -2e^{-x^2} - 2xe^{-x^2}(-2x) \\
 &= -2e^{-x^2} + 4x^2e^{-x^2} \\
 &= 2e^{-x^2}[-1 + 2x^2]
 \end{aligned}$$

$= 0$

DNE

No Ne

$$2e^{-x^2}[-1 + 2x^2] = 0$$

$$2e^{-x^2} = 0$$

No Ne

$$-1 + 2x^2 = 0$$

$$2x^2 = 1$$

$$x^2 = \frac{1}{2}$$

$$x = \frac{1}{\sqrt{2}}$$

$\in \text{Domain}$

$$x = -\frac{1}{\sqrt{2}}$$

$\in \text{Domain}$



$f''(x)$	$(-\infty, -\frac{1}{\sqrt{2}})$	$(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$	$(\frac{1}{\sqrt{2}}, \infty)$
$2e^{-x^2}[-1 + 2x^2]$	+	-	+
	$\cup$	$\cap$	$\cup$

$\therefore f(x)$ is concave up on $(-\infty, -\frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, \infty)$
 $f(x)$ is concave down on $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$$\text{at } x = -\frac{1}{\sqrt{2}} \rightarrow y = e^{-\frac{1}{2}}$$

$$\text{at } x = \frac{1}{\sqrt{2}} \rightarrow y = e^{-\frac{1}{2}} \Rightarrow \infty \left(-\frac{1}{\sqrt{2}}, e^{-\frac{1}{2}} \right)$$

$$\left(\frac{1}{\sqrt{2}}, e^{-\frac{1}{2}} \right)$$

are Inflection points.

key points

intersection point with HA

$$0 = e^{-x^2}$$

None

Random point

at $x=1 \rightarrow y=e^{-1}$
 $(1, e^{-1})$

$$D_f = \mathbb{R}$$

HA $\rightarrow y=0$

$(0,1)$ L.max

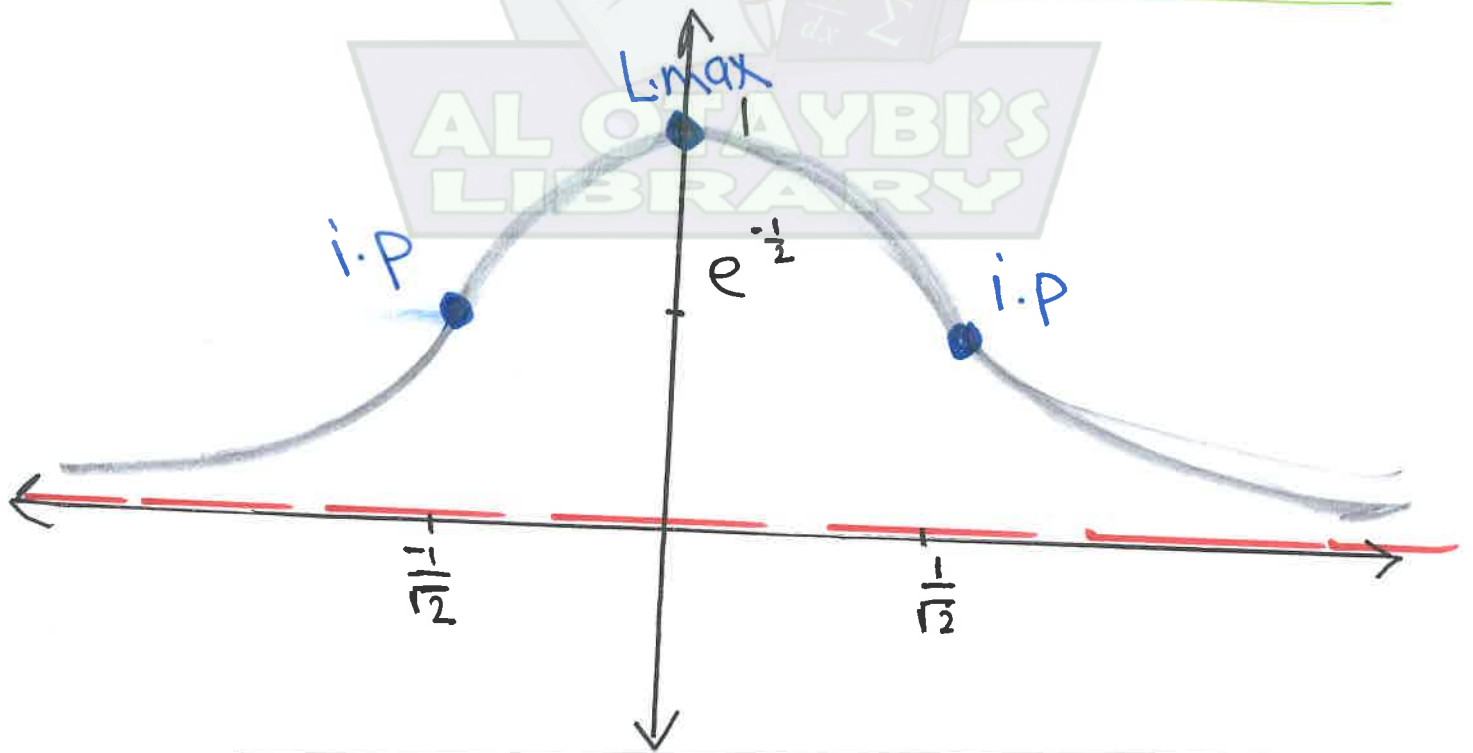
$(1, e^{-1})$

V.A $\rightarrow$ None

$(\frac{-1}{\sqrt{2}}, e^{-\frac{1}{2}}), (\frac{1}{\sqrt{2}}, e^{-\frac{1}{2}})$ i.p's

$\cup (-\infty, \frac{1}{\sqrt{2}}) (\frac{1}{\sqrt{2}}, \infty)$

$\cap (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$



Let $f(x) = \ln(x^2 + 1)$

a) Find the vertical and horizontal asymptotes , if any.

b) Study the symmetry of f

Given that $f'(x) = \frac{2x}{x^2 + 1}$

c) Find the intervals of increasing or decreasing.

d) Find the local maximum and minimum , if any.

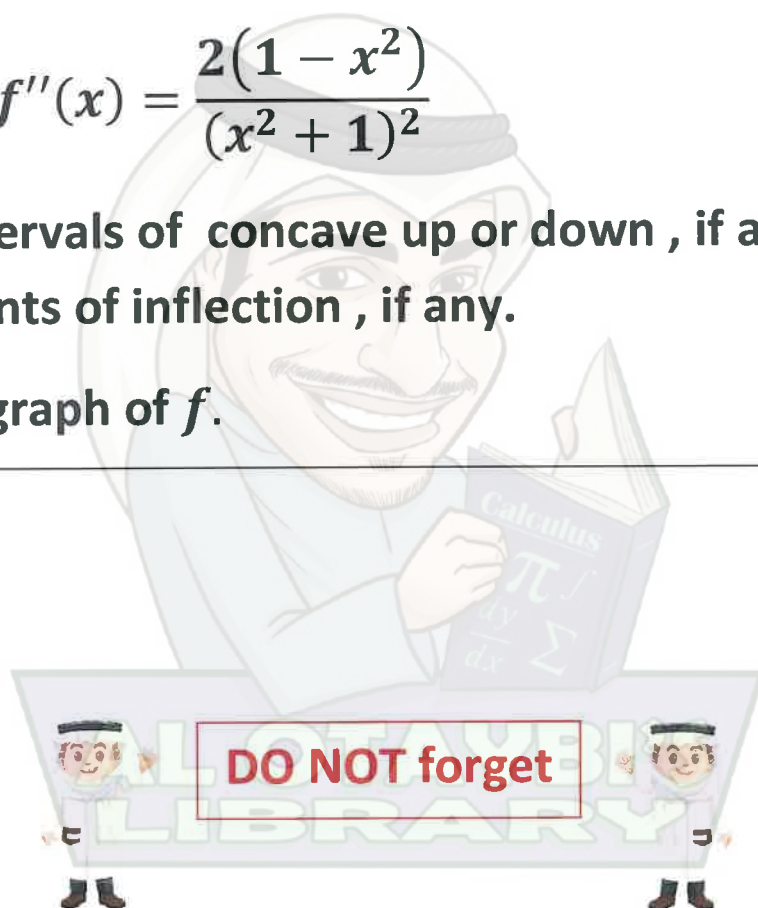
Given that $f''(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}$

e) Find the intervals of concave up or down , if any.

f) Find the points of inflection , if any.

g) Sketch the graph of f .

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On Limit

$$\ln(\infty) = \infty \quad , \quad \ln(0^+) = -\infty \quad , \quad \ln(1) = 0$$

Domain of $f(x)$: $x^2 + 1 > 0$
 $x^2 > -1$ (always)

$\Rightarrow D_f = \mathbb{R}$

Horizontal asymptote

$\lim_{x \rightarrow \pm\infty} \ln(x^2 + 1) = \ln\infty = \infty$

$\Rightarrow \therefore$ No HA

Vertical asymptote

No V.A

$f'(x) = \frac{2x}{x^2 + 1}$

$= 0$

$2x = 0$

$x = 0 \in D_f$

undefined

$x^2 + 1 = 0$

$x^2 = -1$

None



$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{2x}{x^2 + 1}$	-	+
	$\ominus$	$\oplus$

$\therefore$ The interval of decreasing = $(-\infty, 0)$
 No interval of increasing = $(0, \infty)$

No L. max

at $x = 0$

$y = \ln(0 + 1) = \ln 1 = 0$

$\therefore (0, 0)$ L. min

$$f''(x) = \frac{2(1-x^2)}{(x^2+1)^2}$$

$$= 0$$

$$2(1-x^2) = 0$$

$$\div 2 \quad 1-x^2 = 0$$

$$(1-x)(1+x) = 0$$

$$x = -1 \quad x = 1$$

EDP

EDP

undefined

$$x^2 + 1 = 0$$

$$x^2 = -1$$

None



$f''(x)$	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$\frac{2(1-x)(1+x)}{(x^2+1)^2}$	-	+	-
	+	+	+



∴ the interval of concave upward = $(-1, 1)$

the interval of concave downward = $(-\infty, -1)$ and $(1, \infty)$

∴ at $x = -1 \rightarrow y = \ln(1+1) = \ln 2$

at $x = 1 \rightarrow y = \ln(1+1) = \ln 2$

∴ $(-1, \ln 2)$ $(1, \ln 2)$ are inflection points

key points

Random point

at $x=2 \rightarrow y = \ln(4+1) = \ln 5$

$\Rightarrow P(2, \ln 5)$

$D_f = \mathbb{R}$

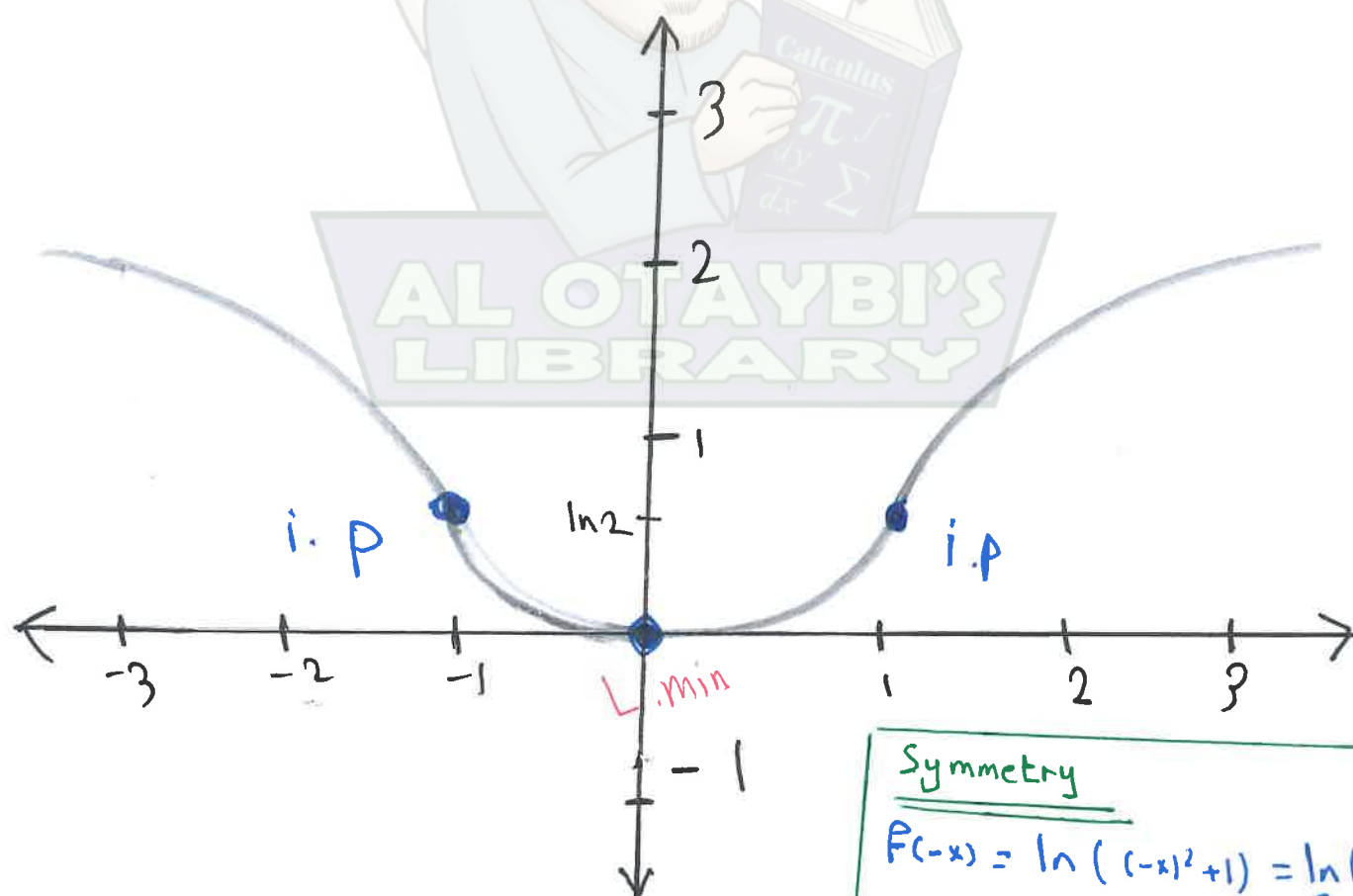
HA $\rightarrow$ None

V.A $\rightarrow$ None

$(0,0)$ L.min

$(-1, \ln 2)$ $(1, \ln 2)$ inflection points.

$\cup$ $(-1, 1)$ $(-\infty, -1)$ $(1, \infty)$
 $(2, \ln 5)$



Symmetry

$$f(-x) = \ln((-x)^2 + 1) = \ln(x^2 + 1) = f(x)$$

$\therefore$ The symmetry about y-axis.

A rational function f has the following properties.

- its domain is $(-\infty, -1) \cup (-1, \infty)$
 - $f(-2) = 0$, $f(0) = 2$, $f(1) = 1$, $f(2) = 3$
 - $\lim_{x \rightarrow -1^-} f(x) = \infty$ and $\lim_{x \rightarrow -1^+} f(x) = \infty$
 - $\lim_{x \rightarrow -\infty} f(x) = -\infty$ and $\lim_{x \rightarrow \infty} f(x) = \infty$
 - f is increasing on $(-\infty, -1)$ and $(1, \infty)$, f is decreasing on $(-1, 1)$.
 - The graph of f is concave upward on $(-\infty, -1)$ and $(-1, 2)$
 - The graph of f is concave downward on $(2, \infty)$
- a) Find the local maximum and minimum of f , if any.
b) Find the point of inflection on the graph , if any.
c) Sketch the graph of f .

Domain of $f(x)$: $\mathbb{R} \setminus \{-1\}$

No H.A ($\lim_{x \rightarrow -\infty} f(x) = -\infty$ and $\lim_{x \rightarrow \infty} f(x) = \infty$)

$x = -1$ is V.A ($\lim_{x \rightarrow -1^-} f(x) = \infty$ and $\lim_{x \rightarrow -1^+} f(x) = \infty$)

$f'(x)$	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
			

$\therefore$ at $x = 1$ $\longrightarrow$ there is Local minimum

$x = 1 \longrightarrow f(1) = 1$

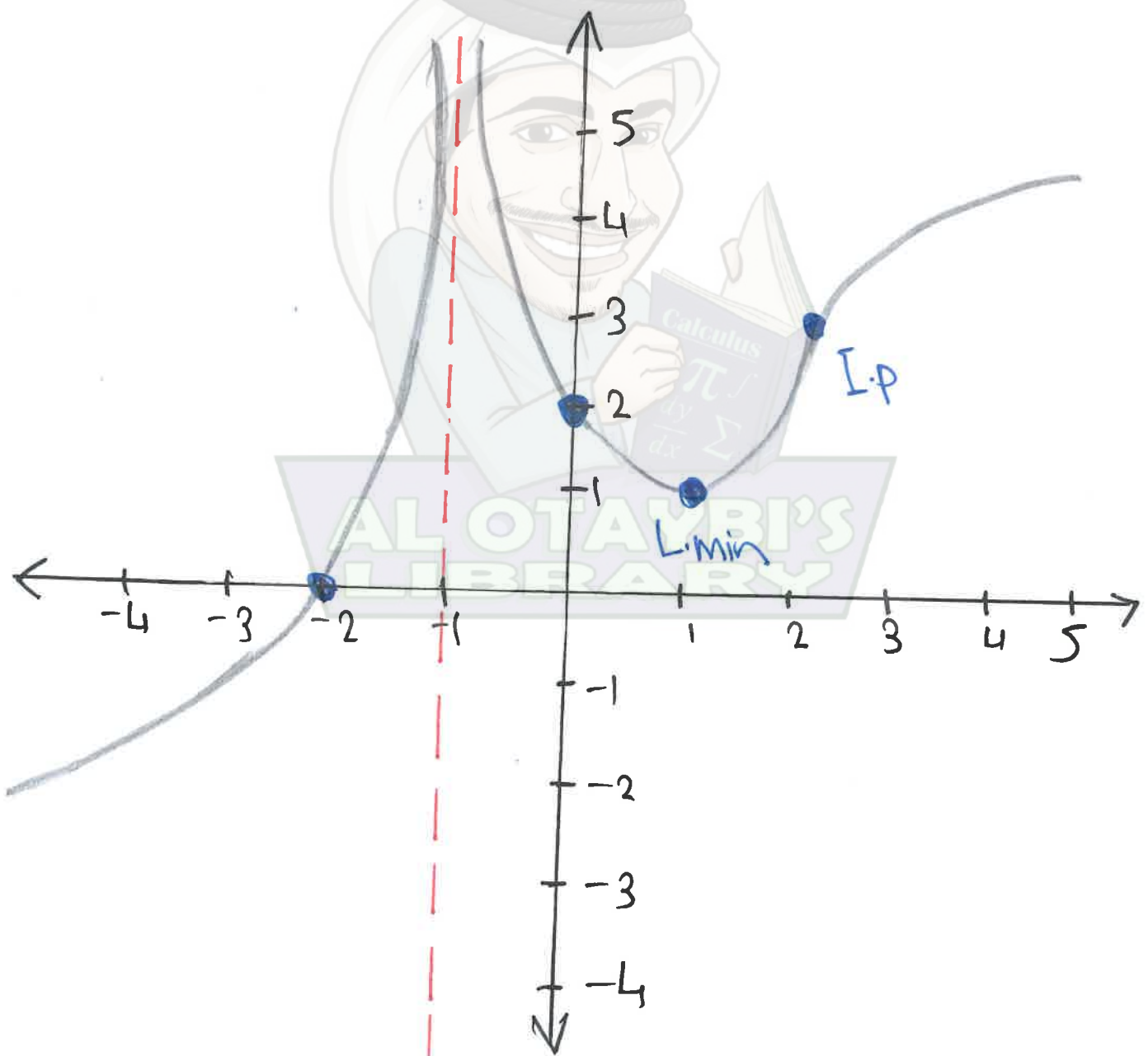
$\therefore$ $(1, 1)$ L.min

$f''(x)$	$(-\infty, -1)$	$(-1, 2)$	$(2, \infty)$
			

$\therefore$ at $x=2 \longrightarrow$ there is inflection point
 $x=2 \longrightarrow f(2) = 3$

$\therefore$ $(2, 3)$ is inflection point

Key points: $(-2, 0)$, $(0, 2)$



Sketch a possible graph of a function f that satisfies the following conditions.

a) $f(0) = 0$, $f(4) = 3$ and $f(6) = 0$

b) $f'(x) < 0$ on $(-\infty, 0)$ and $(4, \infty)$, $f'(x) > 0$ on $(0, 4)$

c) $f''(x) < 0$ on $(-\infty, 0)$ and $(0, 6)$, $f''(x) > 0$ on $(6, \infty)$

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$f'(x)$	$(-\infty, 0)$	$(0, 4)$	$(4, \infty)$
	$\searrow$	$\nearrow$	$\searrow$

$\therefore$ at $x = 0 \longrightarrow f(0) = 0 \longrightarrow \therefore (0, 0)$ is Local minimum
 $\therefore$ at $x = 4 \longrightarrow f(4) = 3 \longrightarrow \therefore (4, 3)$ is Local maximum

$f''(x)$	$(-\infty, 0)$	$(0, 6)$	$(6, \infty)$
	$\cap$	$\cap$	$\cup$

$\therefore$ at $x = 6 \longrightarrow f(6) = 0 \longrightarrow \therefore (6, 0)$ is inflection point

